1821 Widener University

Human-Centered Redesign of a Button Hook for Individuals with Dexterity Impairments

Development, Custom Tooling, and Preliminary Usability Evaluation (n = 30)

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SCAN FOR RESOURCES

Thesis, evaluation, & LinkedIn

1 CLINICAL NEED & AIMS

THE PROBLEM

Reduced hand dexterity from arthritis, stroke, and neuromuscular conditions makes fastening buttons one of the most mechanically demanding activities of daily living (ADLs).

Conventional button hooks use a cylindrical ribbed grip that demands sustained pinch and grip force — exactly the capacity that is impaired — and provide no wrist or palm support.

~21% of U.S. adults live with doctor-diagnosed arthritis.

Independent dressing is among the first ADLs lost — driving loss of autonomy and caregiver burden.

OBJECTIVE

Reduce sustained grip demand and improve button capture reliability through a clinician-informed, human-centered redesign.

AIM 1: HANDLE — Shift Load from Fingers to Palm

- Open contoured palm support + external finger grooves distribute contact force across the hand
- 35°–40° upward curvature promotes a neutral wrist posture during engagement
- Retention configurations compared: fixed 30°, fixed 40°, and pivoting strap
- Pivoting self-cinching Velcro strap + ¼" closed-cell neoprene padding mechanically couple the hand — no active squeezing required

AIM 2: HOOK — Maximize Capture and Retention

- Spring-tempered stainless-steel wire selected to overcome spring-back seen in softer wire
- Loop geometry refined for better capture, threading, retention and release

DESIGN SCREENING

- 7 CAD handle concepts → 4 3D-printed prototypes (A–D) → baseline → final design.
- 5 hook mechanisms evaluated: picker claw, encapsulating capture, cross-linked forceps, pusher-assisted, wire loop. Active mechanisms were eliminated for excess actuation force, added weight, and fabrication/cleaning complexity.

2 ENGINEERING DESIGN, TOOLING & FINAL PROTOTYPE

METHODS: **Commercial-device review + PT/OT clinical consultation → CAD concept generation → FDM (PLA) prototyping → laser-cut custom tooling → iterative expert evaluation → refined integrated device**

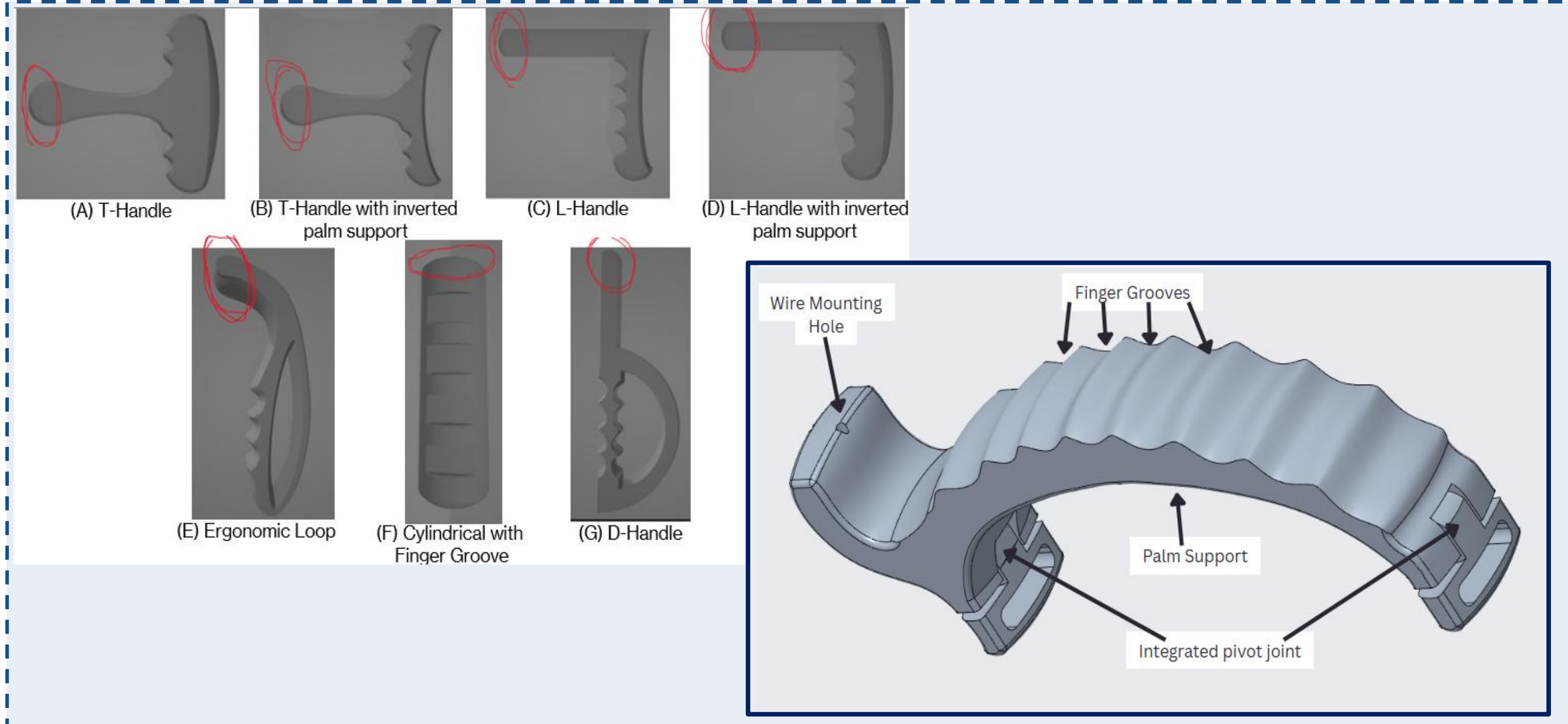


FIGURE A — HANDLE EVOLUTION

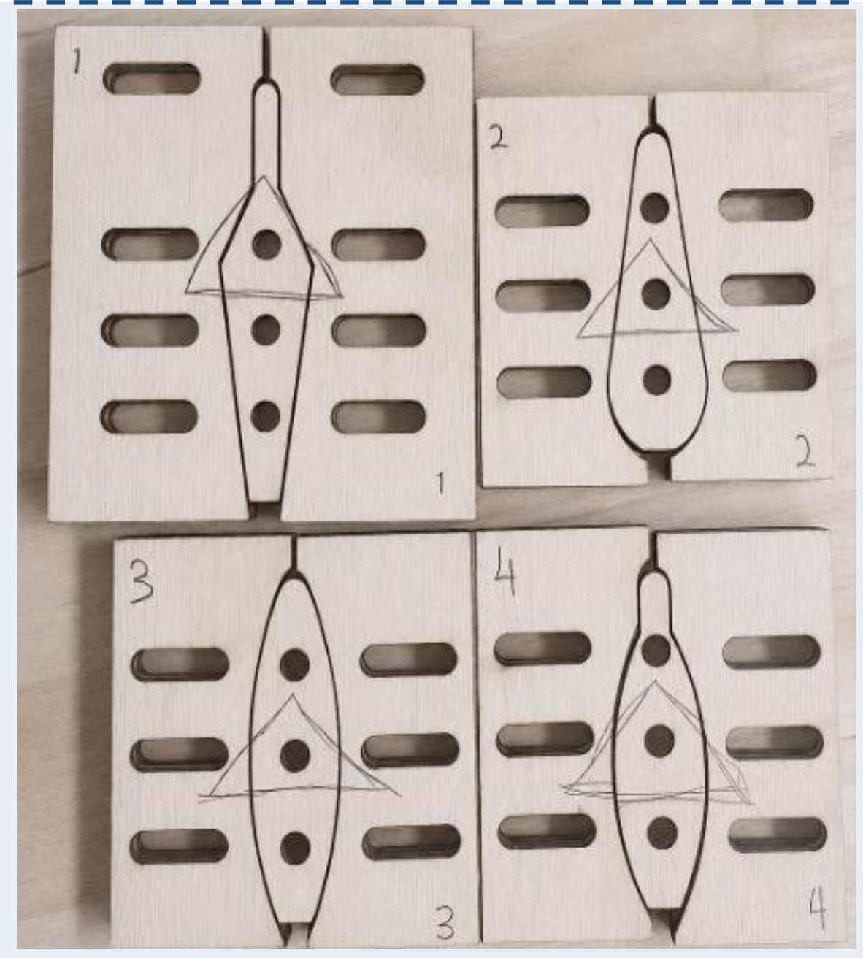


FIGURE B — CUSTOM WIRE-FORMING TOOLING

Final design with enlarged curved palm support with external finger grooves so the palm carries a greater share of the applied force, while the upward curvature drives neutral wrist alignment.

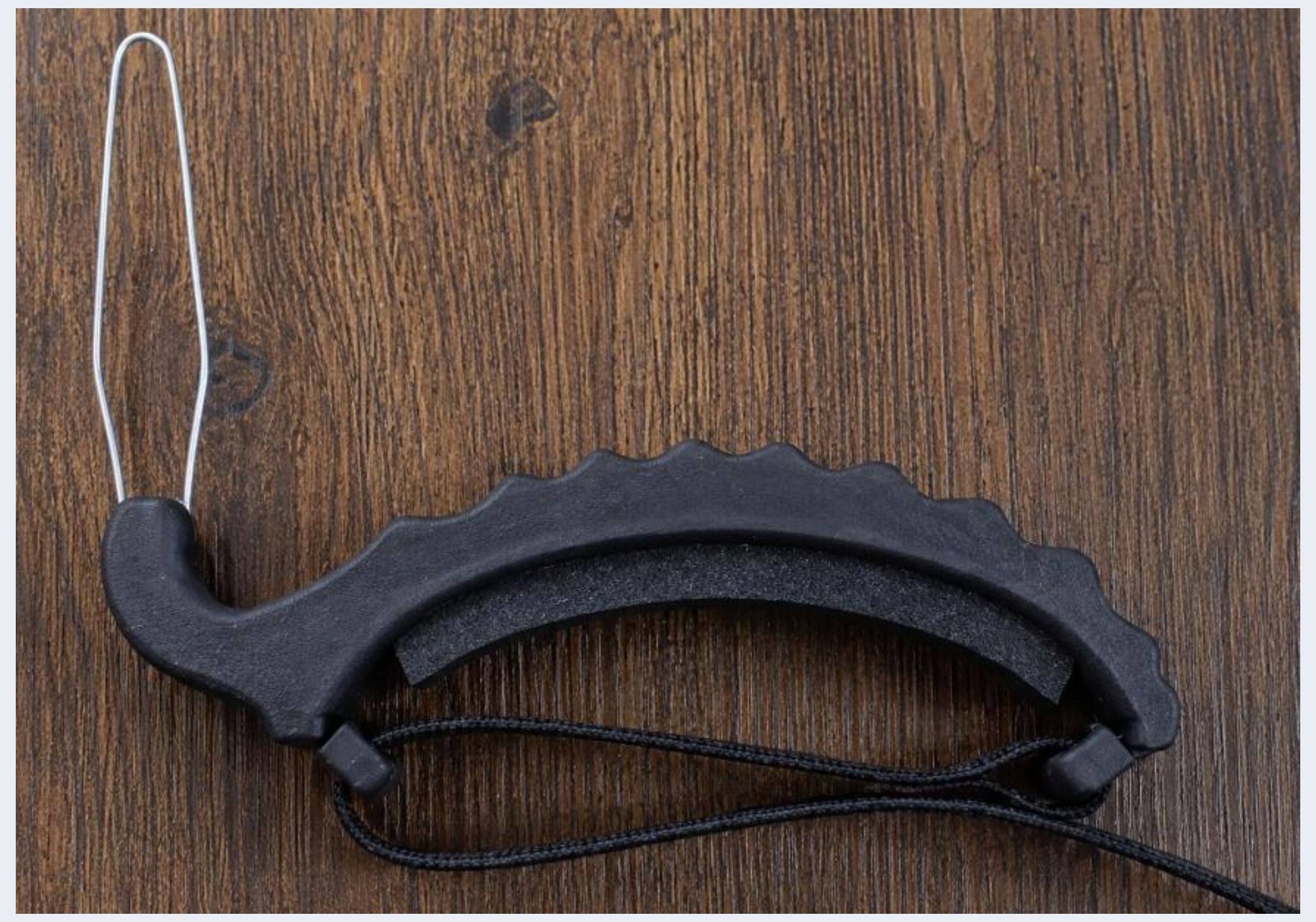


FIGURE C — FINAL INTEGRATED PROTOTYPE

Pivoting strap

Self-cinching Velcro couples the hand without active grip

Palm support + padding

Distributes contact pressure over a larger area

35°–40° offset

Finger grooves and curvature promote neutral wrist

Refined wire loop

Spring-tempered steel; tuned entry aperture & retention

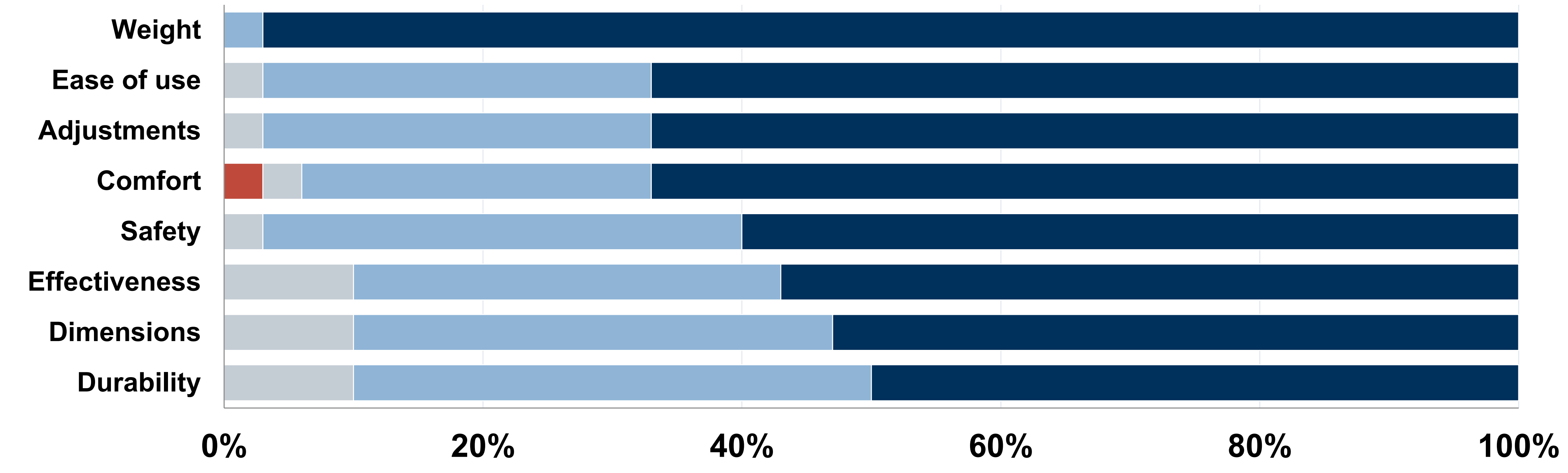
3 USABILITY RESULTS (n = 30)

STUDY DESIGN

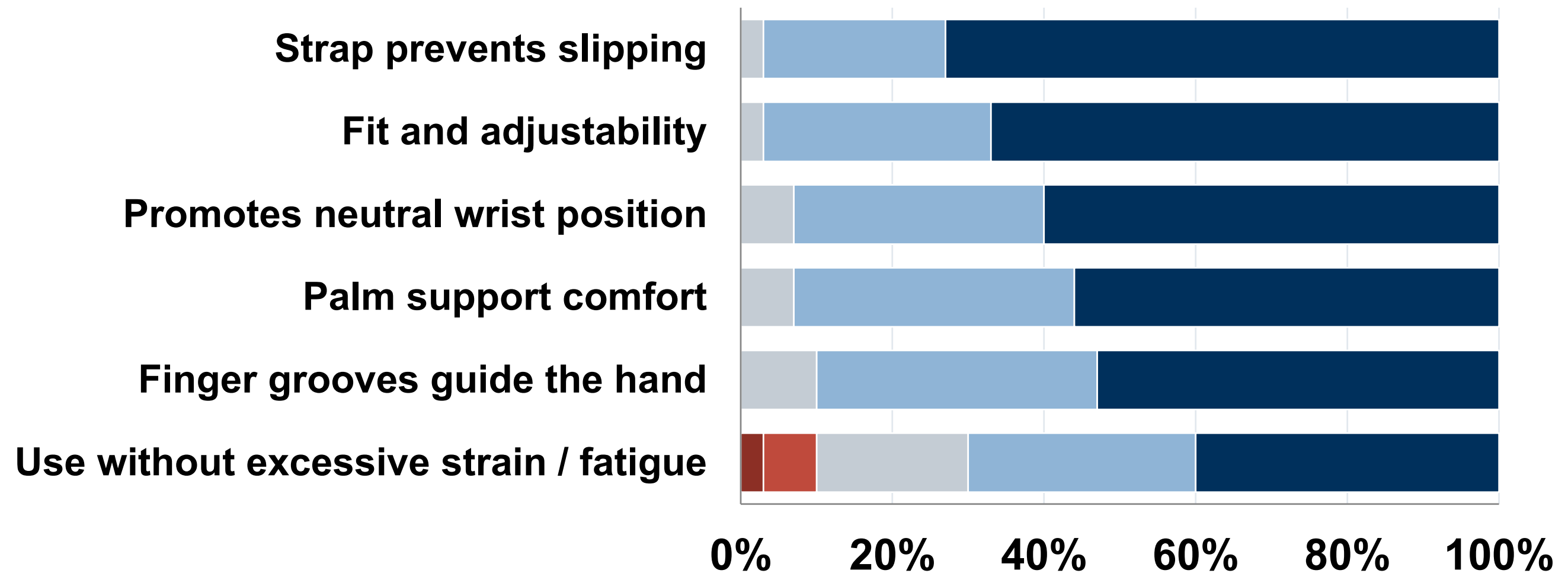
30 graduate PT/OT volunteers · standardized shirt-buttoning task · five-point survey with 8 adapted QUEST 2.0 items plus handle and mechanism items · response frequencies + open-ended thematic analysis.

🟥 Strongly disagree / not satisfied 🟥 Disagree ⬜ Neutral 🟦 Agree / quite satisfied 🟦 Strongly agree / very satisfied

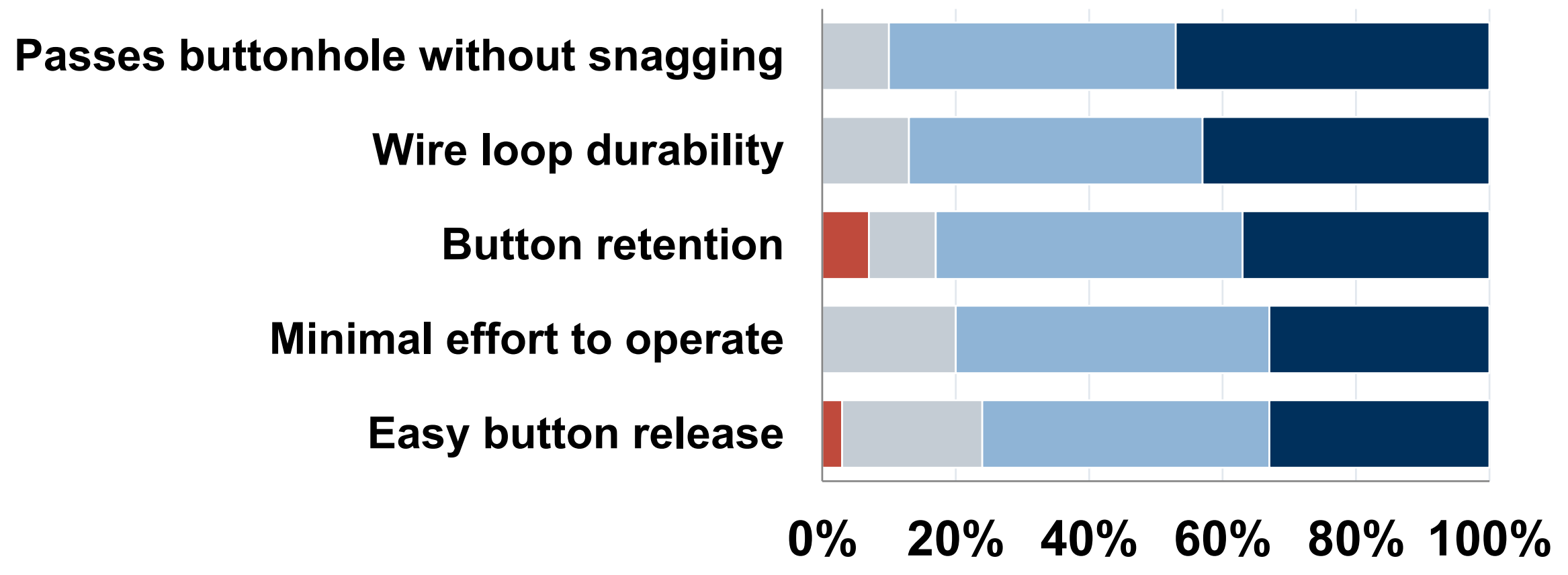
QUEST 2.0 device satisfaction — % of participants (n = 30)



Handle usability — % of participants (n 30)



Mechanism usability — % of participants (n = 30)



FINDINGS & PARTICIPANT FEEDBACK

- Strengths: strap retention, adjustable fit, low weight, perceived wrist comfort.
- Improvement priorities: button retention and release, stronger strap material, broader hand-size adaptability.

LIMITATIONS & NEXT STEPS

- Limitations: healthy graduate PT/OT cohort used as a usability proxy; no diagnosed dexterity impairment; no commercial comparator; self-reported outcomes only.
- Next: evaluate target clinical cohorts (e.g., rheumatoid arthritis, post-stroke); head-to-head benchmarking vs. a commercial button hook; quantify grip effort; palm pressure mapping; motion-capture wrist deviation; task completion time and attempts.

CONCLUSION

A palm-supported, strap-retained handle paired with a jig-formed spring-tempered wire loop delivered strong preliminary usability; button retention and release remain the priority for the next design iteration.

ACKNOWLEDGMENTS

Widener University funding; Lab Engineer; PT/OT faculty, students, and all volunteer participants.